

# Sonder

## Screen fast. Build smart.

Instant EV charger site screening from street-level scans.

Operators evaluate ~50 candidate sites to build one EV charger. Sonder screens them in seconds — using the real-world measurements the city's already been scanned for.

**The whole idea:** the costly part of siting is sending someone to measure each candidate — space, fit, distance to power, what you'd trench through. Cyvl already captured all of that in 3D. We read it, score it, and rank the 50 down to the few worth a real visit.

### THE PITCH — WHAT WE TELL JUDGES

#### The story

**1 • The hook.** "To build one EV charger, operators evaluate dozens of candidate sites — often around fifty. Almost every one needs someone to physically go measure it: is there room, does it meet ADA, how far to the nearest power, what would we have to dig through. That's thousands of dollars and weeks of delay, per site, mostly on sites they'll reject."

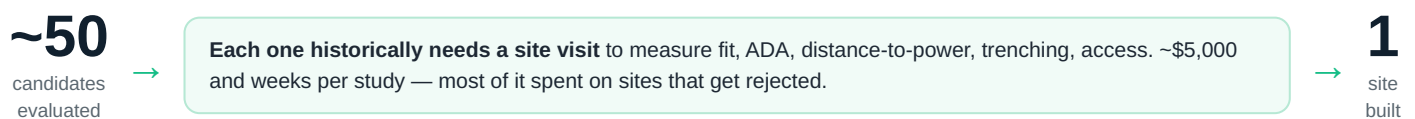
**2 • The insight.** "But all those measurements already exist. Cyvl scanned every street in this city in 3D, to sub-centimeter accuracy. The exact data a site visit produces is sitting in the scan."

**3 • The product.** "Sonder reads it. Type an address and in 60 seconds you get the measurements, a fit-and-feasibility verdict, an estimated connection cost, and a zoning check — the whole desktop screen, automated. The 50 candidates rank instantly; the expensive visit gets spent only on the finalists."

**4 • The close.** "We don't replace the final survey — we kill the screening truck roll. We measured it on real Somerville scans. Every street that gets scanned becomes a city we can screen."

### WHY IT MATTERS

#### The 50-to-1 problem



Sonder compresses the front of that funnel. The physical screen that used to require a truck moves to a 60-second desktop query, so the costly study only happens on the shortlist.

|                                                       |                                        |                                                        |                                                            |
|-------------------------------------------------------|----------------------------------------|--------------------------------------------------------|------------------------------------------------------------|
| <b>~\$5,000</b><br>cost of one site feasibility study | <b>4–6 wks</b><br>turnaround per study | <b>sub-cm</b><br>Cyvl dimensional accuracy (its claim) | <b>\$250k+</b><br>typical cost of one DC fast-charge build |
|-------------------------------------------------------|----------------------------------------|--------------------------------------------------------|------------------------------------------------------------|

A site at 15% utilization loses money; the same hardware at 35% is profitable in three years — and the difference is almost always location. Get the screen right and you stop stranding six-figure builds.

*~50-to-1 is the illustrative screening reality, not a published constant — present it as the typical funnel, validated in customer discovery.*

## The measurements that replace the visit

This is the heart of the product. Every item below comes straight from the Cyvl digital twin — no mock data, no truck roll.

**MEASURED DIRECTLY FROM CYVL**

- **Usable space** + how many charging stalls physically fit
- **ADA clearances** — accessible stall + path width pass/fail
- **Distance to nearest power** (pole / transformer) — the top cost driver
- **Trench path + surface to cut** — asphalt vs. concrete
- **Obstructions** — trees, hydrants, signs, driveways
- **Access & sightlines** — can vehicles get in and out

**PAIRED FROM PUBLIC DATA (one is enough for MVP)**

- **Zoning / land use** — is charging allowed (Somerville open data)
- **Traffic / demand proxy** — MassDOT counts, OSM amenities
- **Competing chargers** ≤3 km — DOE AFDC locator (free)
- **Grid signal** — substation proximity + hosting-capacity map if the utility publishes one **one signal, not the focus**

**FLAG, NEVER FAKE (shown as "verify on site")**

True grid capacity (utility's model) · host willingness · final permit approval · anything underground or behind a fence. Flagging these is credibility — it shows we know the industry.

WHAT THE USER SEES

## The app flow



## The report — the screen judges remember

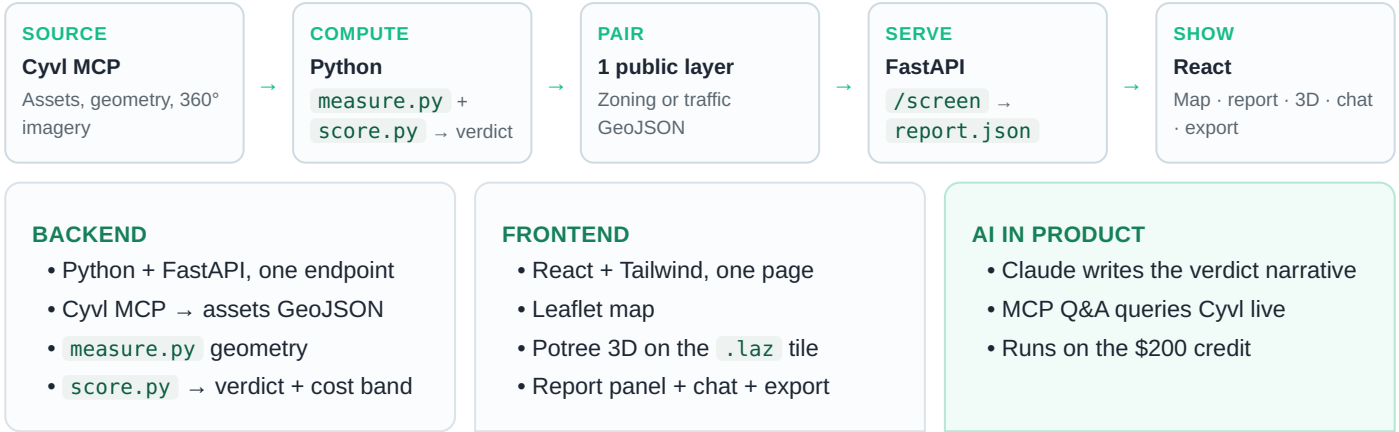
| Block             | Shows                                                      | Source   |
|-------------------|------------------------------------------------------------|----------|
| Verdict           | Go / Conditional / No-go + one-line reason                 | score.py |
| Physical fit ★    | Usable area, stalls that fit, ADA pass/fail, layout sketch | Cyvl     |
| Connection cost ★ | Distance to power, trench length, surface, \$ band         | Cyvl+CV  |
| Zoning            | Permitted? + permit pathway                                | public   |
| Demand            | Traffic proxy, amenities, nearby chargers                  | public   |
| Verify on site    | Grid capacity · host · final permit — flagged              | flag     |

★ The physical measurements + connection-cost estimate are the differentiators no map-based competitor can produce.

HOW IT'S BUILT

# Architecture

**Key decision:** never touch the 500 GB raw cloud. Query the **digital twin via the Cyvl MCP** — assets already carry geometry + attributes. Pull one small `.laz` tile per site only for the 3D visual.



Sponsor tools (use two, genuinely)

| Tool     | How it's wired in                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------|
| NVIDIA   | CV model on Cyvl imagery detects transformers + surface type — inputs to the connection-cost estimate. |
| Autodesk | Documented Cyvl → Recap → Civil 3D path. Auto-export the conceptual site layout.                       |

## The rest of the picture

CUSTOMERS & COMPETITORS

- **Buyers:** CPOs (ChargePoint, EVgo, Electrify America), site-acquisition / EPC firms, property hosts, fleets. ~7 of 80+ operators run 1,000+ sites.
- **Competitors:** EVpin, Sitetracker, GIS tools — all rank demand on a *map*.
- **Our wedge:** we measure the physical site from a real 3D scan. They can't.
- **Pricing:** \$100–500/report · \$2k–10k/mo SaaS · \$100k+/yr API.

3-MINUTE DEMO

- **0:00** "50 sites to build 1, each needs a visit."
- **0:30** Type a real Somerville address. Data pulls live.
- **1:00** Verdict + measurements appear.
- **1:30** Spotlight the connection-cost estimate.
- **2:00** Ask the chat "why rejected?" — answered from Cyvl.
- **2:30** Export CAD. "Same engine, any scanned street."

### One-day build plan

| Block | Goal                                                                                                                  |
|-------|-----------------------------------------------------------------------------------------------------------------------|
| 0–1 h | Connect Cyvl MCP. Pick 2–3 real demo sites. Pull their assets + a <code>.laz</code> tile.                             |
| 1–3 h | Claude Code writes <code>measure.py</code> + <code>score.py</code> . Validate by eye. Lock <code>report.json</code> . |
| 3–4 h | Add zoning/traffic layer. NVIDIA CV for transformer/surface (or hand-label a few).                                    |
| 4–7 h | Claude builds the React app: map, report, 3D tile, chat, CAD export.                                                  |
| 7–8 h | Hardwire the demo sites end-to-end so it can't fail live. Polish. Rehearse.                                           |

# Claude Code — the one prompt to start

Using the Cyvl MCP, query all digital-twin assets within 80 m of {address}: poles, transformers, sidewalk widths, pavement surface, curb geometry. Return GeoJSON + nearest 360 image. Then write `measure.py` (usable area, 2.7x5.5 m stalls that fit, ADA pass/fail, distance to nearest power, trench length, surface crossed) and `score.py` (rules -> Go/Conditional/No-go + cost band). Output `report.json`.

### RISKS ALREADY HANDLED

**500 GB** → query the twin, not raw cloud. **Survey overclaim** → "replaces the screening visit, not the surveyor." **Off-curb parcels** → confirm coverage + pick reachable demo sites. **Idea collision** → lead with physical measurements + connection cost. **Demo risk** → 2–3 hardwired sites.

### BEFORE YOU WRITE CODE

## Data checklist

| Dataset                                                    | Source                         | Status            | MVP? |
|------------------------------------------------------------|--------------------------------|-------------------|------|
| Digital-twin assets (poles, transformers, widths, surface) | Cyvl MCP                       | Have              | MUST |
| Zoning OR traffic layer                                    | Somerville open data / MassDOT | Get it (1 hr)     | MUST |
| .laz tile for 3D visual                                    | Cyvl                           | Have              | MUST |
| Competing chargers                                         | DOE AFDC locator (free)        | Hardcode for demo | Nice |
| Substation / grid signal                                   | OSM / Eversource map if exists | Check             | Nice |

Confirm on the platform first: that the Somerville scan reaches off-curb parcels (not just the road), and that the twin exposes transformer/pole locations + sidewalk widths. Those two checks decide the demo sites.